ASA Seminar, Oct-2025

Optimal control problem with state constraints for cancer chemotherapy and treatment optimization.

D. Lassounon[1,2]

joint with: A. Belmiloudi[1] & M. Haddou[1]

IRMAR-UMR[1] 6625, INSA Rennes, CNRS
CRAN-UMR[2] 7039, Université de Lorraine, CNRS

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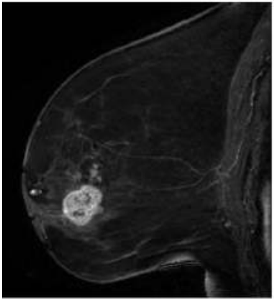
Introduction

A tumor is a group of abnormal cells that form a mass:



Source: Ritse M. Mann *et al.*:Breast MRI:State of the art,2019 and www.passeportsante.net

From (CIRC):

- 484000 new cases in France with 191000 deaths in 2022.
- First country in the world in terms of breast cancer index in 2022.

Introduction

Several therapeutic options are possible: surgery, radiotherapy, chemotherapy, etc.

Introduction 2/2

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- Acute drug toxicity on normal cells.
- Resistance of tumor cells to drugs.
- Tumor resistance after treatments (secondary cancer).

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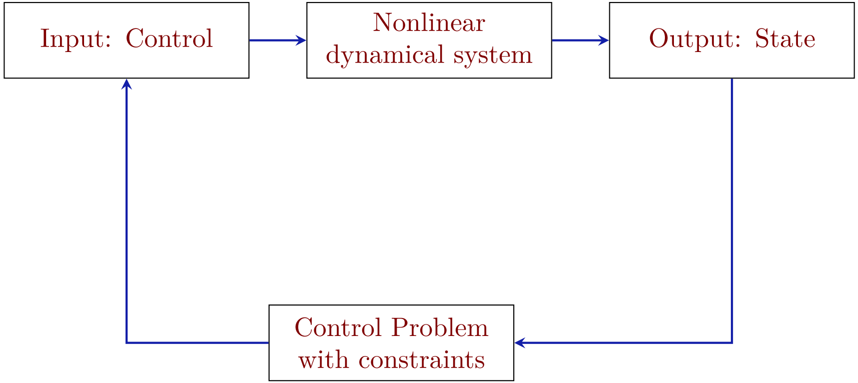
Mathematical Modeling "+" Optimal control problem with state constraint.

Process


Input: Control
Nonlinear dynamical system
Output: State
Control Problem with constraints

State of the art

Mathematical modeling

- T. Yankeelov, *et al*. 2018; A.M. Jarrett , *et al*.2019; D.A. Hormuth, *et al*. 2019; Resende, A., et *al*. 2021; Johnson K., *et al*. 2020; A.M. Jarrett, *et al*. 2020; D.A. Hormuth, *et al*. 2021; ...*etc*

Optimal control

- Segun O., *et al*. 2018; Jarrett A.M., *et al*. 2020; P. Colli, *et al*. 2021; Guillermo L., *et al*. 2021; M. Garavello, *et al*. 2024; ...*etc*

Goal of the work

Eradication of malignant breast tumors by proposing a new methodology:

- using a formalism of the optimal control problem:

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  - a constraint on control to limit the acute toxicity of drugs.

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Eradication of malignant breast tumors by proposing a new methodology:

- using a formalism of the optimal control problem:
  - a constraint on control to limit the acute toxicity of drugs.
  - a constraint on the state to limit the progression of resistant tumors.

Outline of the talk

1. Mathematical modeling
2. Optimal control problem
3. Applications to realistic problems
   - Treatment of breast tumors
   - The influence of state constraint in treatment
   - Treatment of a tumor in a 3D breast domain
   - Treatment of breast tumors in the presence of noise data
   - Treatment of lung tumors
4. Conclusion and future work

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Mathematical model 1/2

$$
\begin{cases}
\dfrac{\partial \mathbf{u}}{\partial t} - \mathrm{div}(\mathrm{D}(t,\mathbf{x})\nabla \mathbf{u}) + \mathrm{K}_1(t,\mathbf{x})\mathrm{h}_1(\mathbf{u}(t,\,\mathrm{x})) - \mathrm{K}_2(t,\mathbf{x})\mathbf{u} + \alpha_0 \phi(t,\mathbf{x})\mathbf{u} = 0 \\
\qquad\qquad\qquad\qquad \text{in } \mathcal{Q} = (0,T)\times\Omega, \\
\qquad\qquad\qquad (\Omega \text{ is the breast or lung domain}) \\
-\mathrm{D}(t,\mathbf{x})\nabla \mathbf{u}\cdot\mathbf{n} - \mathrm{K}_3(t,\mathbf{x})\mathrm{h}_2(\mathbf{u}(t,\,\mathrm{x})) = 0 \text{ on } \Sigma = (0,T)\times\partial\Omega, \\
\qquad\qquad\qquad (\mathbf{n} \text{ is the outward normal to } \partial\Omega) \\
\mathbf{u}(t=0,\mathbf{x}) = u_0(\mathbf{x}) \text{ in } \Omega,
\end{cases}
\tag{1}
$$

under the control constraint

$$
\mathbf{a}(\mathbf{x}) \le \phi(t,\mathbf{x}) \le \mathbf{b}(\mathbf{x}) \quad a.e.(t,\mathbf{x}) \in \mathcal{Q}, \tag{2}
$$

with (for $i = 1,2$)

$$
\mathrm{h}_i : z \in \mathbb{R} \longmapsto \mathrm{h}_i(z) = \left|\begin{array}{ll} z^{\rho_i} & \text{if } \rho_i \in \mathbb{N}\backslash\{1\}, \\ z|z|^{\rho_i-1} & \text{if } \rho_i \in \mathbb{R}\setminus\mathbb{N},\, \rho_i > 1,\ \rho_1 + 1 \ge 2\rho_2. \end{array}\right. \tag{3}
$$

Mathematical model

Model variables	**definitions and dimension**
$\mathbf{u}$	Tumor density *dimensionless*
D	Diffusion function $[mm^2 \cdot d^{-1}]$
K_1	Intrinsic growth rate $[d^{-1}]$
K_2	Proliferation rate $[d^{-1}]$
ϕ	Drug concentration $[\mu\mathrm{M}]$
α_0	Efficacity parameter $[1/d \cdot \mu\mathrm{M}]$
K_3	Migration function $[mm \cdot d^{-1}]$
u_0	Initiale tumor density *dimensionless*
$\mathbf{a}, \mathbf{b}$	Positive functions $[\mu M]$
T	Total time of treatment in days $[d]$

Mathematical analysis 1/2

(H1) : $u_0 \in \mathrm{W}^{r,q}(\Omega)$ *with* $r = 0, 1$ *and* $1 < q < \infty$, $\Omega \subset \mathbb{R}^m \, (m \geq 2)$. *The coefficients* ρ_i *for* $i = 1, 2$ *are fixed such that*

$$(I_1) : \begin{cases} \text{(i) } \textit{if } r = 0, \textit{ then } \rho_1 \in]1; \dfrac{m+2q}{m}] \textit{ and } \rho_2 \in]1; \dfrac{m+q}{m}]. \\ \text{(ii) } \textit{if } r = 1 \textit{ and } q < m, \textit{ then } \rho_1 \in]1; \dfrac{m+q}{m-q}] \textit{ and } \rho_2 \in]1; \dfrac{m}{m-q}]. \\ \text{(iii) } \textit{if } r = 1 \textit{ and } q \geq m, \textit{ then } \rho_i \in]1; \infty[\textit{ for } i = 1, 2. \end{cases}$$

For the parameter of the model, we assume that: $\mu_1 \geq \mathrm{D}(t, \mathrm{x}) \geq \mu_0 > 0$, *and that* $0 < \underline{\mathrm{K}_i} \leq \mathrm{K}_i(t, \mathrm{x}) \leq \overline{\mathrm{K}_i}$ *for all* $(t, \mathrm{x}) \in \bar{\mathcal{Q}}$, *with* $\overline{\mathrm{K}_i}, \underline{\mathrm{K}_i} \in \mathbb{R}_+^*$.

Mathematical analysis 1/2

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(H2) : *The bounds* $\mathbf{a}, \mathbf{b} \in \mathrm{L}^{\mathbf{p}_c}(\Omega)$, $0 \leq \mathbf{a} \leq \mathbf{b}$, *and* $\mathbf{p}_c$ *satisfies*

$$(I_2) : \begin{cases} \text{(i) } \textit{for } r = 0 : \\ \mathbf{p}_c \in]\dfrac{\rho_1}{\rho_1 - 1}; \dfrac{q\rho_1}{\rho_1 - 1}] \textit{ if } \rho_1 \geq q, \textit{ and } \mathbf{p}_c \in [\dfrac{q}{\rho_1 - 1}; \dfrac{q\rho_1}{\rho_1 - 1}] \textit{ if } \rho_1 < q. \\ \text{(ii) } \textit{for } r = 1 \textit{ and } q < m : \\ \mathbf{p}_c \in]\dfrac{mq}{(m-q)(\rho_1 - 1)}; \dfrac{q\rho_1}{\rho_1 - 1}] \textit{ if } \dfrac{m}{m-q} < \rho_1, \textit{ and } \mathbf{p}_c \in [\dfrac{\rho_1}{\rho_1 - 1}; \dfrac{q\rho_1}{\rho_1 - 1}] \textit{ if } \rho_1 \leq \dfrac{m}{m-q}. \\ \text{(iii) } \textit{for } r = 1 \textit{ and } q \geq m, \ \mathbf{p}_c = \dfrac{q\rho_1}{\rho_1 - 1}. \end{cases}$$

We denote by $\mathcal{U}_{ad}$ *the set of admissible controls defined by*

$$\mathcal{U}_{ad} := \Big\{ \varphi \in \mathrm{L}^{\mathbf{p}_c}(\mathcal{Q}) : \mathbf{a}(\mathrm{x}) \leq \varphi(t, \mathrm{x}) \leq \mathbf{b}(\mathrm{x}), a.e.\ t \in [0; T] \Big\}.$$

*We suppose that the assumptions **H1** and **H2** are satisfied.*

Mathematical analysis 2/2

Theorem

1. *For all* $u_0 \in \mathrm{W}^{r,q}(\Omega)$ *(*$r = 0, 1$ *and* $q > 1$*) and* $\varphi \in \mathcal{U}_{ad}$*, there exists* $\underline{\gamma} > 0$*, a time* $\tau_0 > 0$ *and a unique mild solution* $\mathbf{u}$ *on* $[0; \tau_0]$ *to* (1)-(2)-(3)*. Moreover, if* $\rho_1 + 1 > 2\rho_2$ *or* $\rho_1 + 1 = 2\rho_2$ *with* $4\nu_0(q-1)\inf\limits_{\mathcal{Q}} \mathrm{K}_1 \geq c_\Omega(\inf\limits_{\Sigma} \mathrm{K}_3)^2(\rho_2 + q - 1)^2$*, then* $\mathbf{u} \in \mathrm{C}([0,T]; \mathrm{W}^{r,q}(\Omega))$ *is the classical solution globally defined in all time* $t \in (0; T)$ *for* (1)-(2)-(3)*.*

Mathematical analysis 2/2

Theorem

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2. *Let* $\mathbf{u}_1$*, resp.* $\mathbf{u}_2$ *the classical solution with respect to* $(u_{0,1}, \varphi_1)$*, resp.* $(u_{0,2}, \varphi_2)$ *belonging to* $\mathrm{W}^{r,q}(\Omega) \times \mathcal{U}_{ad}$*. Then, there exists* $C > 0$ *such that for all* $t \in (0; T)$*,* $\theta \in [0; \underline{\gamma}]$*,* $\epsilon \in [0; \theta]$ *the following holds*

$$
\begin{aligned}
\|\mathbf{u}_1(t,\cdot) - \mathbf{u}_2(t,\cdot)\|_{\mathrm{W}^{2\theta+r,q}(\Omega)} &\leq C t^{-\theta} \Big(\|u_{0,1} - u_{0,2}\|_{\mathrm{W}^{r,q}(\Omega)} + \|\varphi_1 - \varphi_2\|_{\mathcal{U}_{ad}} \Big), \\
\|\frac{\partial \mathbf{u}_1}{\partial t}(t,\cdot) - \frac{\partial \mathbf{u}_2}{\partial t}(t,\cdot)\|_{\mathrm{W}^{-1,q'}(\Omega)} &\leq C t^{\epsilon-\theta} \Big(\|u_{0,1} - u_{0,2}\|_{\mathrm{W}^{r,q}(\Omega)} + \|\varphi_1 - \varphi_2\|_{\mathcal{U}_{ad}} \Big).
\end{aligned}
\tag{4}
$$

3. *For* $u_0 \in \mathrm{W}^{r,q}(\Omega)$ *such that* $0 \leq u_0 \leq c_s$ *and* $\varphi \in \mathcal{U}_{ad}$*, the solution* $\mathbf{u}$ *of* (1)-(2)-(3) *is bounded in* $\mathcal{Q}$ *and verifies*

$$
0 \leq \mathbf{u}(t,\cdot) \leq c_s, \text{ for all } t \in (0;T) \text{ and a.e. in } \Omega, \tag{5}
$$

where $c_s = \left(\dfrac{\overline{\mathrm{K}_2}}{\underline{\mathrm{K}_1}} \right)^{\frac{1}{\rho_1 - 1}}$.

Outline of the talk

In this part, we restrict ourselves to some realistic situations and consider only cases: $\rho_1 \geq 2$, $\rho_2 = 2$, $r = 1$, $q = 2$, $\theta = 0$, $m \leq 3$, *and* $\mathbf{p}_c \geq 2$ *(the existence of such* $\mathbf{p}_c$ *is garanted when* $\rho_1 \geq 2$*).*

Let $\mathcal{K} := \mathrm{C}([0;T])_-$ *be the cone of the negative continuous functions on* $[0;T]$.

Let define the solution operator
$\mathcal{S} : \phi \in \mathcal{U}_{ad} \longmapsto \mathcal{S}(\phi) \in \mathbf{U} := \mathcal{W} \cap \mathrm{C}([0,T];\mathrm{H}^1(\Omega))$ *such that* $\mathcal{S}(\varphi) = \mathbf{u}$ *solution of* (1)-(2)-(3), *where* $\mathcal{W} := \mathrm{L}^2(0,T;\mathrm{H}^1(\Omega)) \cap \mathrm{H}^1(0,T;(\mathrm{H}^1(\Omega))')$.

The cost functional:

$$\mathcal{J}(\phi) = \frac{1}{2}\int_{\mathcal{Q}} \mathbf{u}^2 d\mathbf{x}dt + \frac{\lambda}{\mathbf{p}_c}\int_{\mathcal{Q}} |\phi|^{\mathbf{p}_c} d\mathbf{x}dt, \tag{6}$$

$$G(\phi) = \|\mathbf{u}(t,\cdot)\|^2_{\mathrm{L}^2(\Omega)} - \xi \leq 0, \tag{7}$$

s.t., $\mathbf{u} = \mathcal{S}(\phi)$.

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The control problem:

$$(\mathbf{CP}) \begin{cases} \inf\limits_{\phi\in\mathcal{U}_{ad}} \mathcal{J}(\phi); \\ s.t., \quad G(\phi) \in \mathcal{K}. \end{cases} \tag{8}$$

PART 1.

Theorem

1. *The control problem* $(\mathbf{CP})$ *admits at least one optimal solution* $\phi_* \in \mathcal{U}_{ad}$.
2. *The solution operator* $\mathcal{S}$ *is continuously differentiable from* $\mathcal{U}_{ad}$ *to* $\mathbf{U}$ *such that the derivative* $\mathcal{S}'(\phi) : \varphi \longmapsto w = \mathcal{S}'(\phi) \cdot \varphi$ *at point* $\phi \in \mathcal{U}_{ad}$ *is the unique solution be in* $\mathcal{W} \cap \mathrm{C}([0;T], \mathrm{L}^2(\Omega))$ *in the weak formulation sense of the following problem*

$$\begin{cases} \dfrac{\partial w}{\partial t} - \operatorname{div}(\mathrm{D}\nabla w) + \Big(\mathrm{K}_1 \mathrm{h}_1'(\mathbf{u}) - \mathrm{K}_2 + \alpha_0 \phi\Big) w = -\alpha_0 \varphi \mathbf{u} \ \textit{in} \ \mathcal{Q}, \\ -\mathrm{D}\nabla w \cdot \mathbf{n} - \mathrm{K}_3 \mathrm{h}_2'(\mathbf{u}) w = 0 \ \textit{on} \ \Sigma, \\ w(0,\, \mathrm{x}) = 0 \ \textit{in} \ \Omega. \end{cases} \tag{9}$$

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Moreover, we have the estimates $(\forall\phi \in \mathcal{U}_{ad})$:

$$\|\mathcal{S}(\phi+\varphi) - \mathcal{S}(\phi) - \mathcal{S}'(\phi)\cdot\varphi\|^2_{\mathcal{W}\cap\mathrm{C}([0,T];\mathrm{L}^2(\Omega))} \leq C\|\varphi\|^4_{\mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})}, \tag{10}$$

where φ *is such that* $\phi + \varphi \in \mathcal{U}_{ad}$.

$$\|\mathcal{S}'(\phi_1)\cdot\varphi - \mathcal{S}'(\phi_2)\cdot\varphi\|^2_{\mathcal{W}\cap\mathrm{C}([0,T];\mathrm{L}^2(\Omega))} \leq C\|\phi_1 - \phi_2\|^2_{\mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})}\|\varphi\|^2_{\mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})} \tag{11}$$

for all $(\phi_1, \phi_2, \varphi) \in \mathcal{U}_{ad} \times \mathcal{U}_{ad} \times \mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})$.

Theorem

1. *The state constrainte (7) is qualified if the following condition holds:*

$$\begin{cases} \text{There exists } \phi \in \mathcal{U}_{ad} \text{ such that} \\ \varpi(t) := \mathcal{G}(\phi_*)(t) + \int_\Omega \mathbf{u}_*(t,\, \mathrm{x}) w_\Phi(t,\, \mathrm{x})\, \mathrm{d}\,\mathrm{x} < 0, \forall t \in (0;T) \\ \mathbf{u}_* = \mathcal{S}(\phi_*) \text{ and } w_\Phi \text{ solution of (9) for } \Phi = \phi - \phi_* \;. \end{cases} \tag{12}$$

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2. *Let $\phi_* \in \mathcal{U}_{ad}$ a solution of $(\mathbf{CP})$ and $\mathbf{u}_* = \mathcal{S}(\phi_*)$ such that the condition (12) holds. Then there exist Lagrange multipliers $(\tilde{\mathbf{u}}_*, \mu_*) \in \mathrm{L}^2(0,T;\mathrm{H}^1(\Omega)) \cap \mathrm{L}^\infty(0,T;\mathrm{L}^2(\Omega)) \times BV(0,T)_+$, such that $\mu_* \neq 0$, and that $\tilde{\mathbf{u}}_* = \tilde{\mathcal{S}}(\phi_*)$ is the solution of the following adjoint problem*

$$\begin{cases} -\dfrac{\partial \tilde{\mathbf{u}}_*}{\partial t} - \mathrm{div}(\mathrm{D}\nabla\tilde{\mathbf{u}}_*) + (\mathrm{K}_1\mathrm{h}_1'(\mathbf{u}_*) - \mathrm{K}_2 + \alpha_0\phi_*)\tilde{\mathbf{u}}_* = \mathbf{u}_* + 2\mathbf{u}_*\,\mathrm{d}\mu_*(\mathrm{t}) \text{ in } \mathcal{Q}, \\ -\mathrm{D}\nabla\tilde{\mathbf{u}}_* \cdot \mathbf{n} - \mathrm{K}_3\mathrm{h}_2'(\mathbf{u}_*)\tilde{\mathbf{u}}_* = 0 \text{ on } \Sigma, \\ \tilde{\mathbf{u}}_*(t = T, \cdot) = 0 \text{ in } \Omega. \end{cases} \tag{13}$$

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The multiplier μ_* verifies

$$\mathcal{G}(\phi_*) \in \mathcal{K};\ \int_0^T \mathcal{G}(\phi_*)\,\mathrm{d}\mu_*(t) = 0;\ \mu_* \in \mathcal{K}^-;\ \mu_*(T) = 0, \tag{14}$$

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and we have the following optimality condition

$$\langle \lambda |\phi_*|^{\mathbf{p}_c - 1} - \alpha_0 \mathbf{u}_* \tilde{\mathbf{u}}_*; \phi - \phi_* \rangle_{\mathrm{L}^{\mathbf{p}_c'}(\mathcal{Q}), \mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})} \geq 0 \text{ for all } \phi \in \mathcal{U}_{ad}\, . \tag{15}$$

The penalized cost functional:

$$\mathcal{J}_p(\phi) = \mathcal{J}(\phi) + \frac{\beta_1}{\mathbf{p}_c}\int_{\mathcal{Q}} (\mathbf{a} - \phi)_+^{\mathbf{p}_c} \,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t + \frac{\beta_2}{\mathbf{p}_c}\int_{\mathcal{Q}} (\phi - \mathbf{b})_+^{\mathbf{p}_c} \,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t + \frac{\beta_3}{4}\int_0^T (\mathcal{G}(\phi))_+^2 \,\mathrm{d}t \quad (16)$$

where for all real function u we denote $u_+ = \max(u, 0)$.

The penalized cost functional:

$$\mathcal{J}_p(\phi)=\mathcal{J}(\phi)+\frac{\beta_1}{\mathbf{p}_c}\int_{\mathcal{Q}}(\mathbf{a}-\phi)_+^{\mathbf{p}_c}\,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t+\frac{\beta_2}{\mathbf{p}_c}\int_{\mathcal{Q}}(\phi-\mathbf{b})_+^{\mathbf{p}_c}\,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t+\frac{\beta_3}{4}\int_0^T(\mathcal{G}(\phi))_+^2\,\mathrm{d}t \quad (16)$$

where for all real function u we denote $u_+ = \max(u, 0)$.

We obtained a control problem without constraint:

$$(\mathbf{PCP}):\inf_{\phi\in \mathrm{L}^{\mathbf{p}_c}(\mathcal{Q})}\mathcal{J}_p(\phi). \quad (17)$$

Proposition

- *$\mathcal{J}_p$ is continuously differentiable and its gradient is given by*

$$\nabla \mathcal{J}_p(\phi) = -\alpha_0 \mathbf{u}\tilde{\mathbf{u}} + \lambda\phi^2 - \beta_1(\mathbf{a}_- \phi_*)_+^2 + \beta_2(\phi_* - \mathbf{b})_+^2. \tag{18}$$

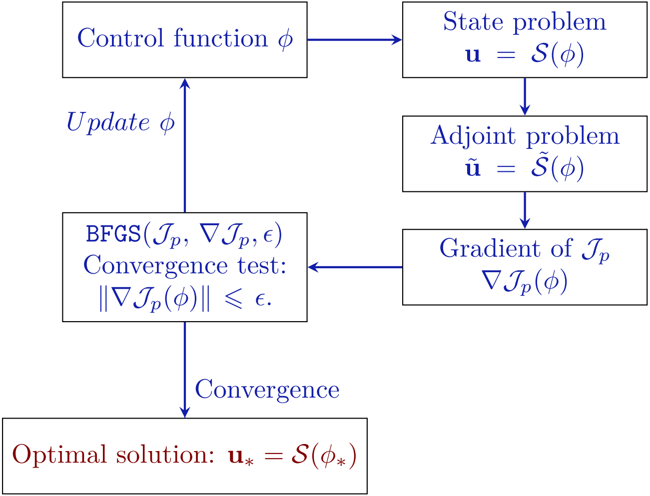
- *Let ϕ_* be a solution of* (**PCP**) *and $\mathbf{u}_* = \mathcal{S}(\phi_*)$ the associated optimal state. Then $\nabla \mathcal{J}_p(\phi_*) = 0$, (since without constraints) i.e,*

$$\lambda\phi_*^2 - \alpha_0 \mathbf{u}_*\tilde{\mathbf{u}}_* - \beta_1(\mathbf{a}_- \phi_*)_+^2 + \beta_2(\phi_* - \mathbf{b})_+^2 = 0, \tag{19}$$

where $\tilde{\mathbf{u}}_*$ verifies the following adjoint system

$$\begin{cases} -\dfrac{\partial \tilde{\mathbf{u}}_*}{\partial t} - \mathrm{div}(\mathrm{D}\nabla\tilde{\mathbf{u}}_*) + (\mathrm{K}_1 \mathrm{h}_1'(\mathbf{u}_*) - \mathrm{K}_2 + \alpha_0\phi_*)\tilde{\mathbf{u}}_* = \mathbf{u}_* + \beta_3 \mathbf{u}_* \mathrm{G}(\phi_*)_+ \ \mathit{in}\ \mathcal{Q}, \\ -\mathrm{D}\nabla\tilde{\mathbf{u}}_* \cdot \mathbf{n} - \mathrm{K}_3 \mathrm{h}_2'(\mathbf{u}_*)\tilde{\mathbf{u}}_* = 0 \ \mathit{on}\ \Sigma, \\ \tilde{\mathbf{u}}_*(t = T, \cdot) = 0 \ \mathit{in}\ \Omega. \end{cases} \tag{20}$$

PART 1. Optimization Algorithm



PART 2. Applications to realistic problems

We consider: $\rho_1 = 3,\ \rho_2 = 2,\ \mathbf{p}_c = \dfrac{2\rho_1}{\rho_1 - 1} = 3,\ r = 1,\ q = 2,\ m = 2$.

Outline of the talk

PART 2-1. Treatment of breast tumors

$$\mathrm{D} = \kappa_0 e^{-\lambda_0 \sigma}; \quad \sigma = \sigma_0 e^{-\delta_0 |\mathbf{x}|^2 t}; \quad \mathrm{K}_1 = \frac{k_1 - k_2 e^{-\gamma_0 \sigma}}{e^{-\gamma_0 \sigma}(1 - e^{-2\gamma_0 \sigma})}; \quad \mathrm{K}_2 = \theta^2 \Psi_1;$$

$$\mathbf{a}(\mathbf{x}) = 0; \quad \mathbf{b}(\mathbf{x}) = \gamma_0 e^{-\gamma_1 [(x-a)^2 + (y-b)^2]}; \quad \mathrm{K}_3 = k_3 \Psi_{1/\Gamma}; \quad \kappa_0 = d_0 \eta_1;$$

$$\eta_1 = 1 + \varsigma_1 \cos(\varsigma_2 \eta_2); \quad \eta_2 = \varsigma_3 \arctan\left(\frac{y - y_0}{\varsigma_4 + \sqrt{(x - x_0)^2 + (y - y_0)^2}} \right);$$

$$\xi = \|u_0\|^2_{\mathrm{L}^2(\Omega)}.$$

Parameters	**Values**
d_0	$1.5\ mm^2 \cdot d^{-1}$
σ_0	18 kPa
λ_0	2.10^{-3} kPa^{-1}
k_2	$2.0\ d^{-1}$
k_1	$4.0\ d^{-1}$
k_3	$3.0\ mm$:
δ_0	$5.55 \cdot 10^{-5}\ mm^{-2} \cdot d^{-1}$
α_0	$1/d \cdot \mu\mathrm{M}$
T	20 days
γ_0	$100\ \mu\mathrm{M}$
γ_1	$20\ mm^2$
θ	2.236 (dimensionless)
ς_1	0.75
ς_2	50
ς_3	2
ς_4	10^{-4}

We used some realistic data from Lorenzo, G., *et al*, . 2021.

PART 2-1. Treatment of breast tumors

(1) Case of a single tumor

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} \text{ in } \Omega, \tag{21}$$

the density of tumor cells centered in (x_0, y_0).

The discretization and mesh informations:

Parameters	**Value**	**Definitions**
Δt	0.2 days (d)	Time step
$\mathbf{x}$	$(x; y)$	Space position
Δx	$1\ mm$	Space discretizations (x)
Δy	$1\ mm$	Space discretizations (y)
n_e	3184	Total mesh elements
n_p	1653	Total mesh points
$(x_0; y_0)$	$(10/3; 10/3)$	Tumor center
$(a; b)$	$(2.9; 10/3)$	
ε	0.02	
$\mathcal{P}^1$		Finite elements (Freefem++)

PART 2-1. Treatment of breast tumors

(1) Case of a single tumor

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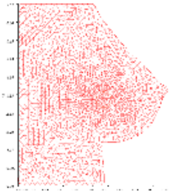
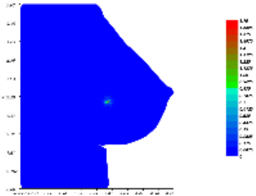
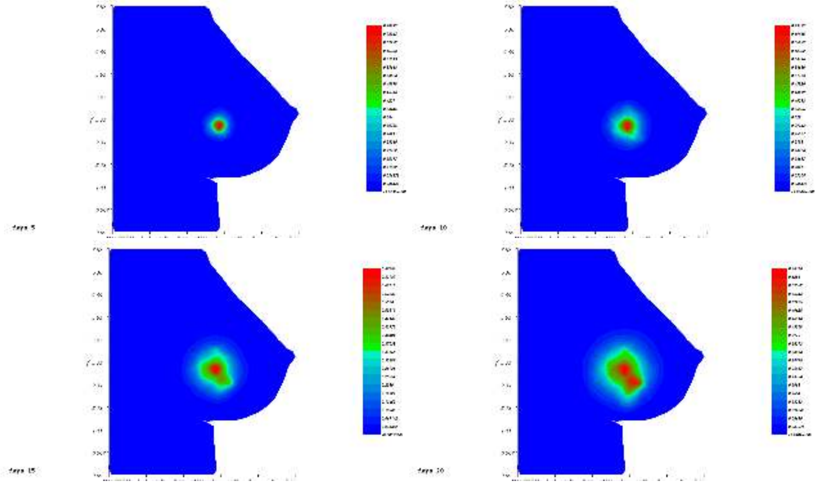




PART 2-1. Treatment of breast tumors

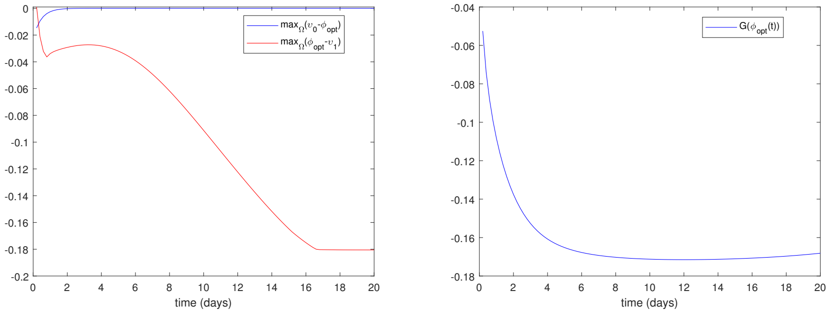
The density of tumor cells $\mathbf{u}$ without treatment *i.e*, $\phi = 0$ in (1) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



PART 2-1. Treatment of breast tumors

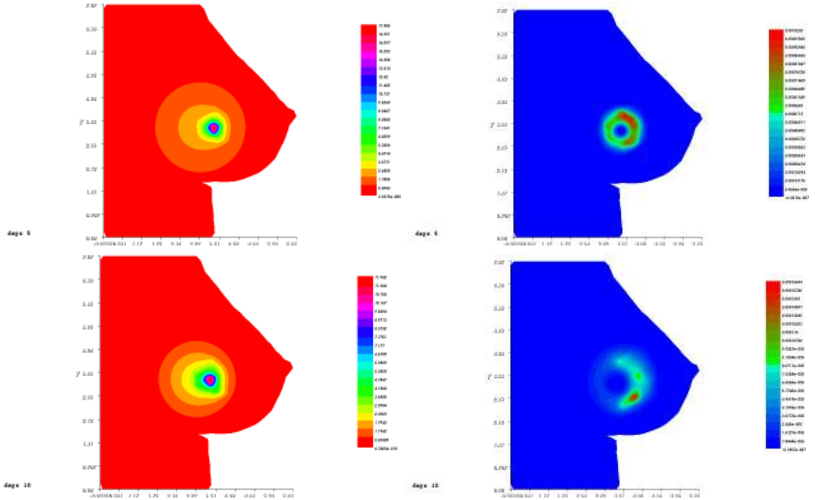
The cost parameters
$\lambda = 0.5 \cdot 10^{-2}$; $\beta_1 = 10^3$; $\beta_2 = 2 \cdot 10^4$; $\beta_3 = 10^6$.

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

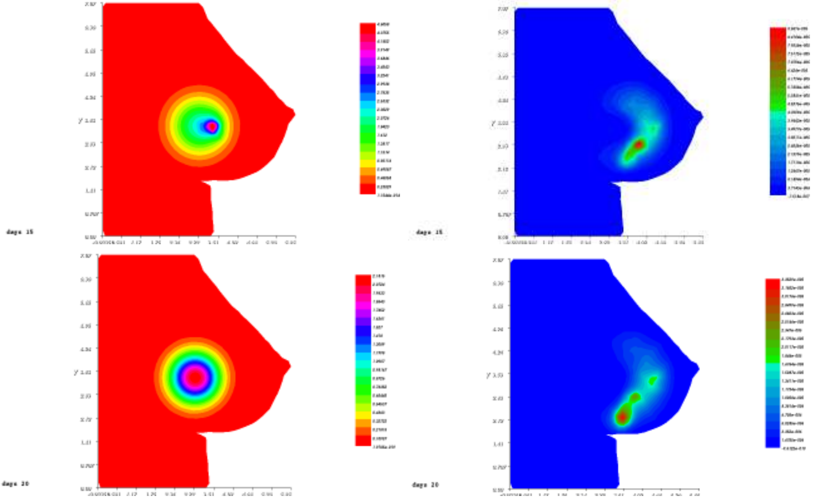


PART 2-1. Treatment of breast tumors

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



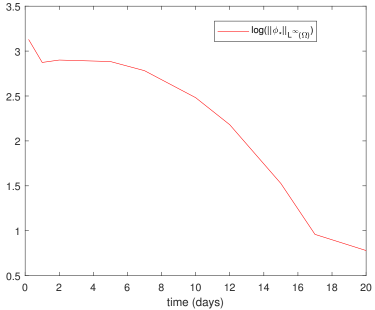
PART 2-1. Treatment of breast tumors

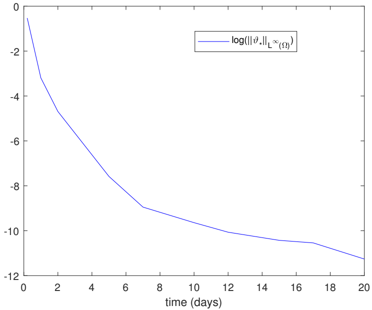


Time (t)	5^{th} days	10^{th} days	15^{th} days	20^{th} days
$\sup_{\Omega} \mathbf{u}_*$	0.00010	$6.5 \cdot 10^{-6}$	$4.96 \cdot 10^{-6}$	$3.35 \cdot 10^{-6}$
$\inf_{\Omega} \mathbf{u}_*$	$6.13 \cdot 10^{-8}$	$2.36 \cdot 10^{-8}$	$3.2 \cdot 10^{-8}$	$9.51 \cdot 10^{-11}$

PART 2-1. Treatment of breast tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





PART 2-1. Treatment of breast tumors

(2) Case of two tumors

$$\mathbf{u}_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} + \eta_1 e^{-\frac{1}{\varepsilon}[(x-\tilde{x}_0)^2+(y-\tilde{y}_0)^2]}, \text{ in } \Omega, \tag{22}$$

the tumor cell density centered at (x_0, y_0) and $(\tilde{x}_0, \tilde{y}_0)$.

The discretization and mesh informations

Parameters	**Value**	**Definitions**
n_e	3310	Total mesh elements
n_p	1716	Total mesh points
$(x_0; y_0)$	$(10/3; 2.12)$	First Tumor center
$(\tilde{x}_0; \tilde{y}_0)$	$(4.03; 2.3)$	Second Tumor center
$(a; b)$	$(3.68; 2.3)$	
ε	0.02	

PART 2-1. Treatment of breast tumors

(2) Case of two tumors

$$\mathbf{u}_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} + \eta_1 e^{-\frac{1}{\varepsilon}[(x-\tilde{x}_0)^2+(y-\tilde{y}_0)^2]}, \text{ in } \Omega, \quad (22)$$

the tumor cell density centered at (x_0, y_0) and $(\tilde{x}_0, \tilde{y}_0)$.

The discretization and mesh informations

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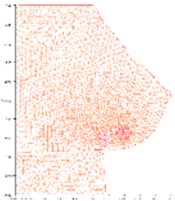
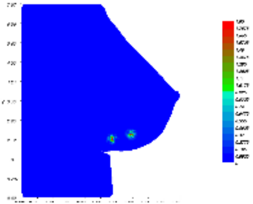
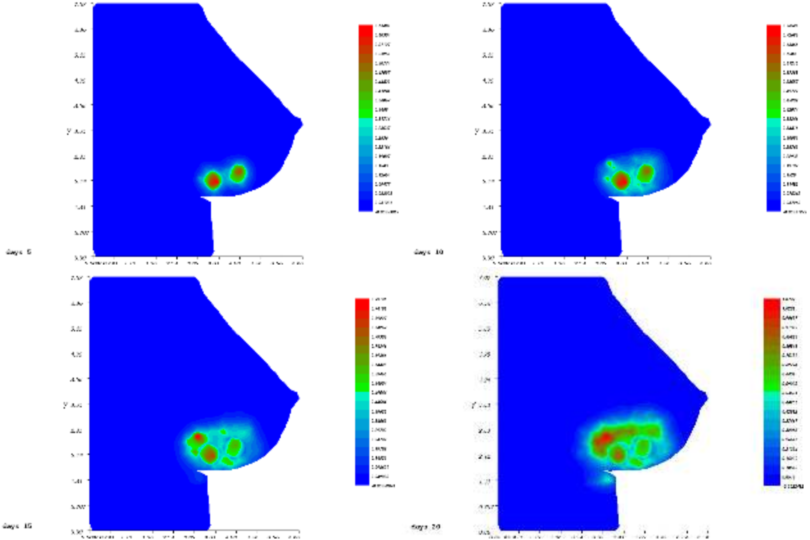




PART 2-1. Treatment of breast tumors

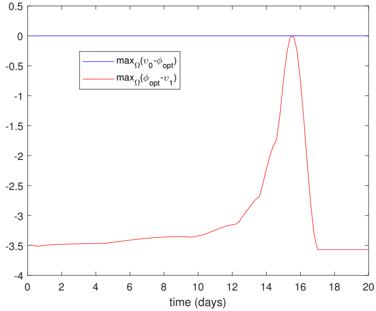
The density of tumor cells $\mathbf{u}$ without treatment *i.e*, $\phi = 0$ in (1) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

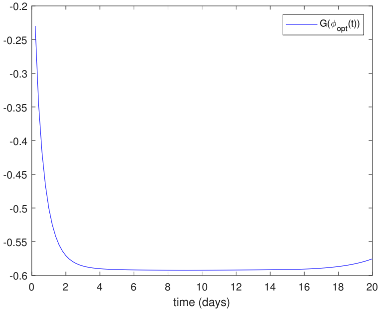


PART 2-1. Treatment of breast tumors

The cost parameters
$\lambda = 10^{-2};\ \beta_1 = 2\cdot 10^3;\ \beta_2 = 10^4;\ \beta_3 = 3\cdot 10^6.$

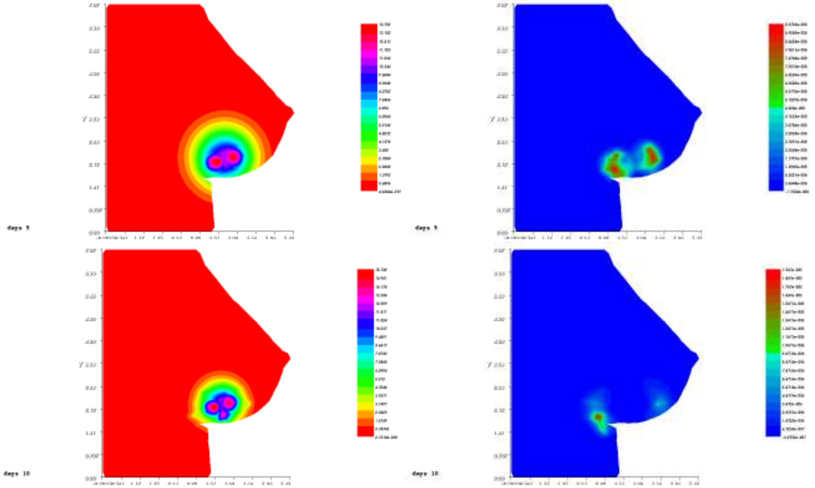
Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf



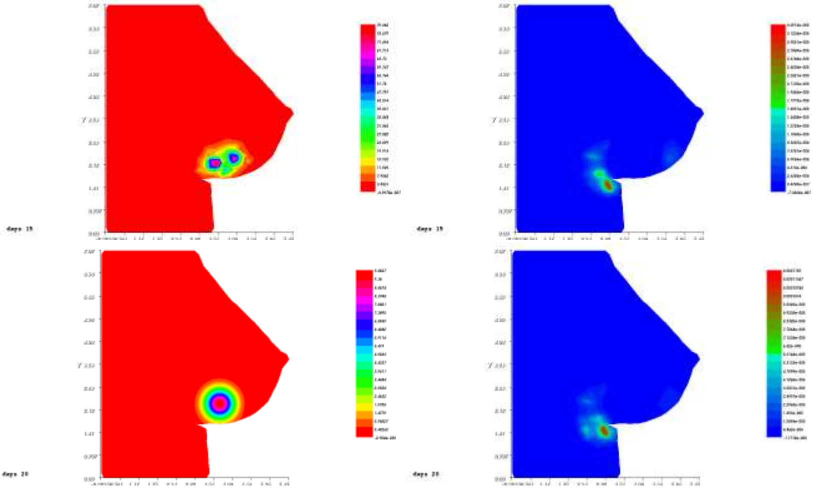


PART 2-1. Treatment of breast tumors

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



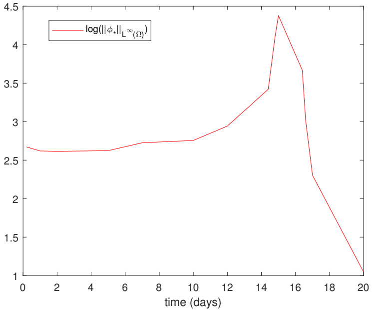
PART 2-1. Treatment of breast tumors

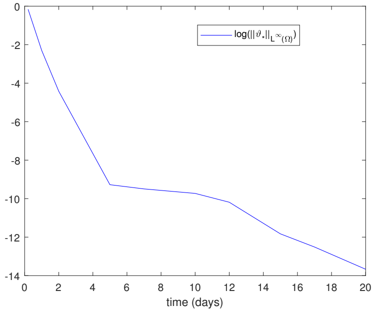


Time (t)	5^{th} days	10^{th} days	15^{th} days	20^{th} days
$\sup_{\Omega} \mathbf{u}_*$	0.00029	$5.22 \cdot 10^{-5}$	$3.34 \cdot 10^{-5}$	$2.77 \cdot 10^{-5}$
$\inf_{\Omega} \mathbf{u}_*$	$3.77 \cdot 10^{-7}$	$1.03 \cdot 10^{-7}$	$2.32 \cdot 10^{-7}$	$3.7 \cdot 10^{-7}$

PARTT 2-1. Treatment of breast tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





PART 2-2. The influence of state constraint

The cost functional:

$$\mathcal{J}(\phi) = \frac{1}{2}\int_{\mathcal{Q}} \mathbf{u}^2 d\mathbf{x}dt + \frac{\lambda}{3}\int_{\mathcal{Q}} |\phi|^3 d\mathbf{x}dt, \tag{23}$$

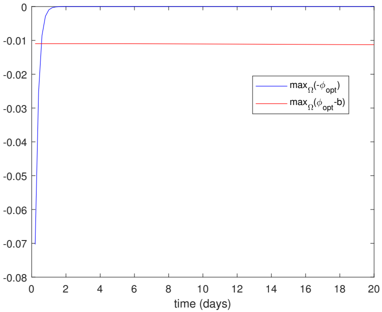
The control problem

$$\begin{aligned} &\inf_{\phi\in\mathcal{U}_{ad}} \mathcal{J}(\phi); \\ &s.t., \mathbf{u} = \mathcal{S}(\phi). \end{aligned} \tag{24}$$

PART 2-2. The influence of state constraint

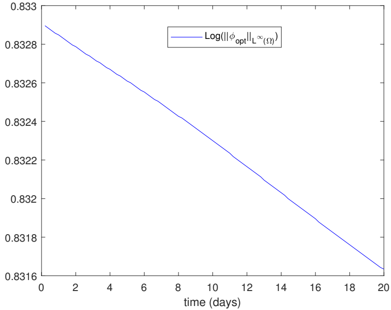
The cost parameters
$\lambda = 10^{-3}$, $\beta_1 = 5 \cdot 10^4$, and $\beta_2 = 10^3$.

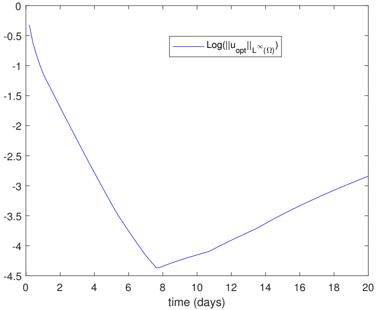
Constraints satisfaction of the optimal control ϕ_* in the following



PART 2-2. The influence of state constraint

Evolution of the maximum value of φ_* on the left and $\mathbf{u}_*$ on the right in the logarithm scale.

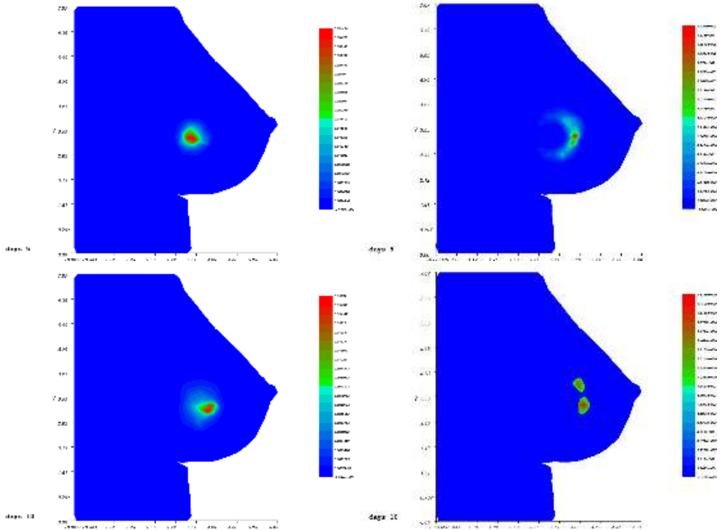




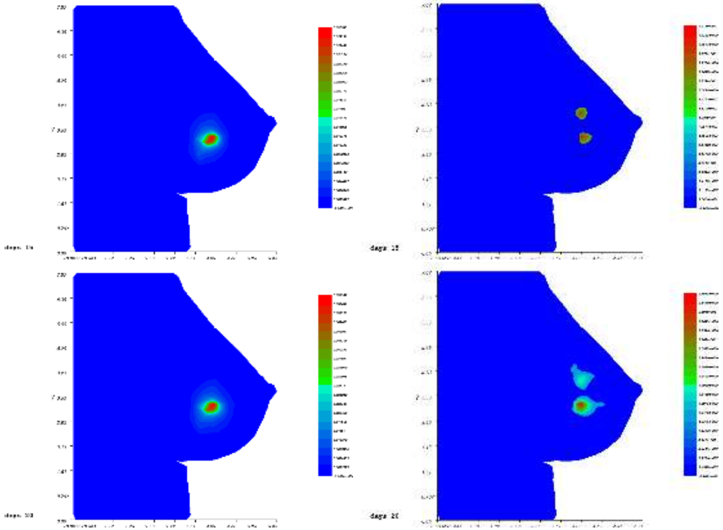
Time (t)	$5th$ days	$10th$ days	$15th$ days	$20th$ days
$\max_{\Omega} \mathbf{u}_*$	0.036	0.016	0.031	0.058
$\min_{\Omega} \mathbf{u}_*$	$4.52 \cdot 10^{-6}$	$1.13 \cdot 10^{-7}$	$1.11 \cdot 10^{-7}$	$8.86 \cdot 10^{-8}$

PART 2-2. The influence of state constraint

The optimal tumor density without state constraint on the left and the optimal tumor density with state constraint on the right.



PART 2-2. The influence of state constraint



PART 2-3. $3D$ Numerical simulations

$$u_0(\mathbf{x}) = ((x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2)e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]} \text{ in } \Omega,$$

The discretization and mesh informations

Parameters	**Value**	**Definitions**
$\mathbf{x}$	(x, y, z)	$3D$ space position
Δx	$1\ mm$	Space Discretizations (x)
Δy	$1\ mm$	Space Discretizations (y)
Δz	$1\ mm$	Space Discretizations (z)
n_e	52980	Total Mesh Elements
n_p	10384	Total Mesh Points
$(x_0; y_0; z_0)$	$(2.34; 3.54; -1.75)$	Tumor center
$(x_1; y_1; z_1)$	$(2.3; 3.40; -1.70)$	

PART 2-3. $3D$ Numerical simulations

$$u_0(\mathbf{x}) = ((x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2)e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2+(z-z_0)^2]} \text{ in } \Omega,$$

The discretization and mesh informations

Parameters	**Value**	**Definitions**
$\mathbf{x}$	(x, y, z)	$3D$ space position
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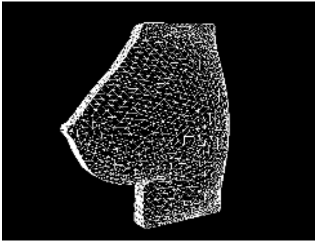
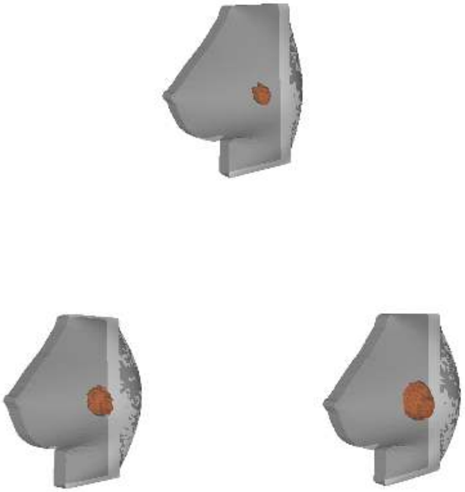


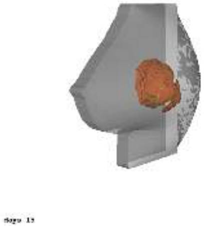
PART 2-3. $3D$ Numerical simulations

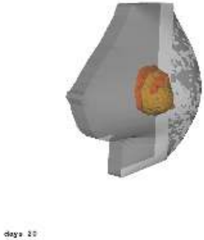
The density of tumor cells $\mathbf{u}$ without treatment *i.e*, $\phi = 0$ in (1) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



days 5

PART 2-3. $3D$ Numerical simulations

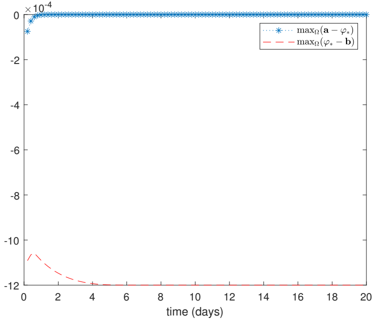


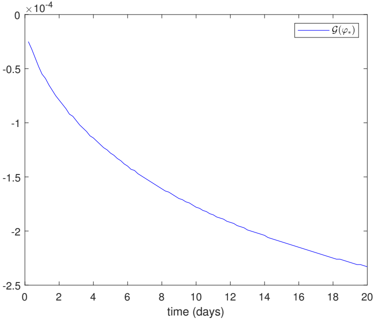


PART 2-3. $3D$ Numerical simulations

The penalized cost parameters
$\lambda = 10^{-2}$; $\beta_1 = 10^5$; $\beta_2 = 5 \cdot 10^4$; $\beta_3 = 10^4$.

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf





PART 2-3. $3D$ Numerical simulations

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

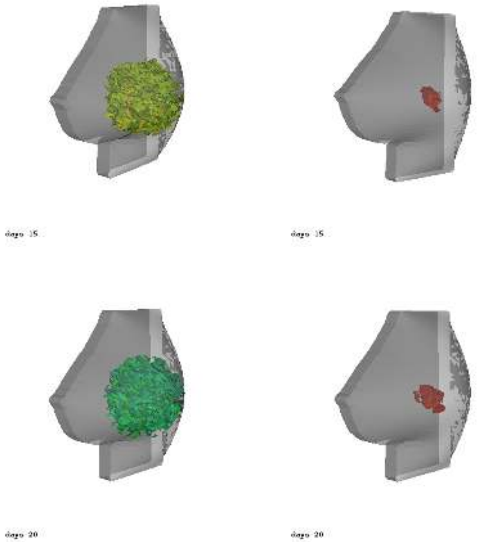
days 5

days 5

days 10

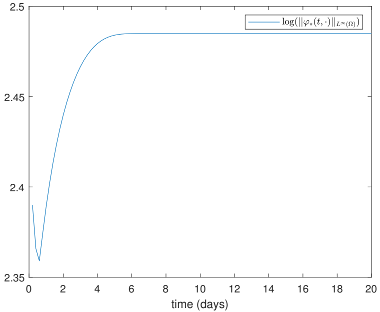
days 10

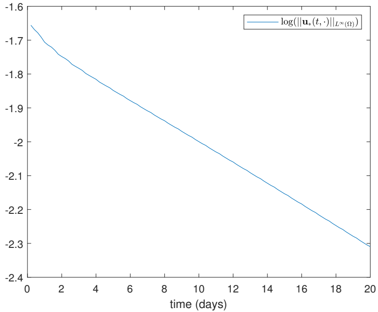
PART 2-3. $3D$ Numerical simulations



PART 2-3. $3D$ Numerical simulations

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





Time (t)	$5th$ days	$10th$ days	$15th$ days	$20th$ days
$\max\limits_{\Omega} \mathbf{u}_*$	0.01587	0.01362	0.01160	0.009130
$\min\limits_{\Omega} \mathbf{u}_*$	0.003261	0.004813	0.005233	0.005079

PART 2-4. Stochastic mathematical model

$$
\begin{cases}
\mathrm{d}\mathbf{u} = \left(\mathrm{div}(\mathrm{D}(t,\, \mathrm{x};\omega)\nabla\mathbf{u}) + \mathrm{K}_2(t,\, \mathrm{x};\omega)\mathbf{u} - \mathrm{K}_1(t,\, \mathrm{x};\omega)\mathbf{u}^3\right)\mathrm{d}t \\
\qquad\qquad + \sigma_0 \mathrm{K}_3(t,\, \mathrm{x};\omega)\mathbf{u}\,\mathrm{d}\mathrm{W}_1(t) - \alpha_0\phi(t,\, \mathrm{x};\omega)\mathbf{u}\,\mathrm{d}\mathrm{W}_2(t) \\
\qquad\qquad\qquad\qquad \text{in } (0,T)\times\mathcal{O}\times\Omega, \\
\mathrm{D}(t,\, \mathrm{x};\omega)\nabla\mathbf{u}\cdot\mathbf{n} = 0 \text{ on } \Sigma\times\Omega, \\
\mathbf{u}(0,\, \mathrm{x};\omega) = u_0(\mathrm{x},\omega) \text{ in } \mathcal{O}\times\Omega,
\end{cases}
\tag{25}
$$

under the pointwise constraint

$$
0 \le \varphi(t,\, \mathrm{x};\omega) \le \mathbf{b}(\mathrm{x},\omega),\ a.e.\ (t,\, \mathrm{x};\omega)\in\mathcal{Q}\times\Omega. \tag{26}
$$

PART 2-4. Stochastic control problem

The cost functional:

$$\mathcal{J}(\varphi) := \mathbb{E}_{\Omega}\left(\frac{1}{2}\int_{\mathcal{Q}} |\mathbf{u}(t,\, \mathrm{x};\omega)|^2 \,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t + \frac{\lambda}{3}\int_{\mathcal{Q}} |\phi(t,\, \mathrm{x};\omega)|^3 \,\mathrm{d}\,\mathrm{x}\,\mathrm{d}t\right). \tag{27}$$

PART 2-4. Stochastic control problem

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The control problem **(CP)**:

$$\begin{aligned} &\inf_{\phi\in\mathcal{U}_{ad}} \mathcal{J}(\phi); \\ &\text{s.t. } \mathcal{G}(\mathbf{u}) = \|\mathbf{u}(t,\cdot)\|^2_{\mathrm{L}^2(\mathcal{O})} - \zeta \le 0,\ a.s., \end{aligned} \tag{28}$$

where

$$\mathcal{U}_{ad} = \{\phi \in \mathrm{L}^3(\Omega;\mathrm{L}^3(\mathcal{Q})),\ 0 \le \varphi(t,\,\mathrm{x};\omega) \le \mathbf{b}(\,\mathrm{x},\omega), a.e.\ t\in[0,T], a.s.\omega\in\Omega\}.$$

PART 2-4. Preliminary results

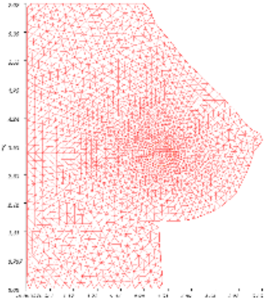
Let $\mathbf{U}$ be a uniform random distribution on $[0,1]$, for $(t,\,\mathrm{x}) \in (0,T)\times\bar{\mathcal{O}}$:

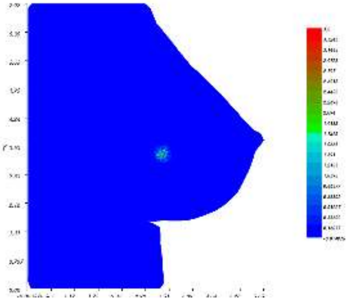
$$\mathrm{D} = \mathrm{D}_0 e^{-\lambda_0\sigma}\cdot\mathbf{U};\quad \sigma = \varrho_0 e^{-\delta_0|\mathbf{x}|^2 t};\quad \mathrm{K}_1 = \frac{k_1 - k_2 e^{-\gamma_0\sigma}}{e^{-\gamma_0\sigma}(1-e^{-2\gamma_0\sigma})}\cdot\mathbf{U};\quad \mathrm{K}_2 = \theta^2\mathrm{K}_1;$$

$$\mathrm{K}_3 = k_3\cdot\mathbf{U};\quad \mathrm{D}_0 = d_0\eta_1;\quad \mathbf{b} = \gamma_0 e^{-\gamma_1[(x-x_1)^2+(y-y_1)^2]}\cdot\mathbf{U};\quad \zeta = \|u_0\|^2_{\mathrm{L}^2(\mathcal{O}\times\Omega)}$$

$$\xi_1 := \frac{\mathrm{dW}_1}{\mathrm{d}t} = (1+\cos(\eta_2))\frac{\mathrm{e}^{-(\frac{t}{\tau})^2}}{\tau\sqrt{2\pi}};\quad \xi_2 = \sigma_0\xi_1.$$

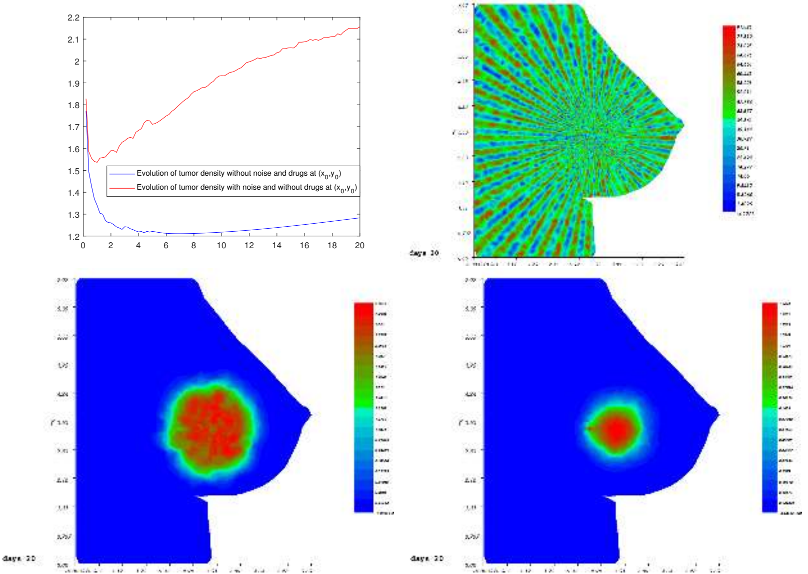
We consider the initial tumor density $u_0(\,\mathrm{x},\omega) = 2\eta_1\,\mathrm{e}^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]}\cdot\mathbf{U}(\omega)$





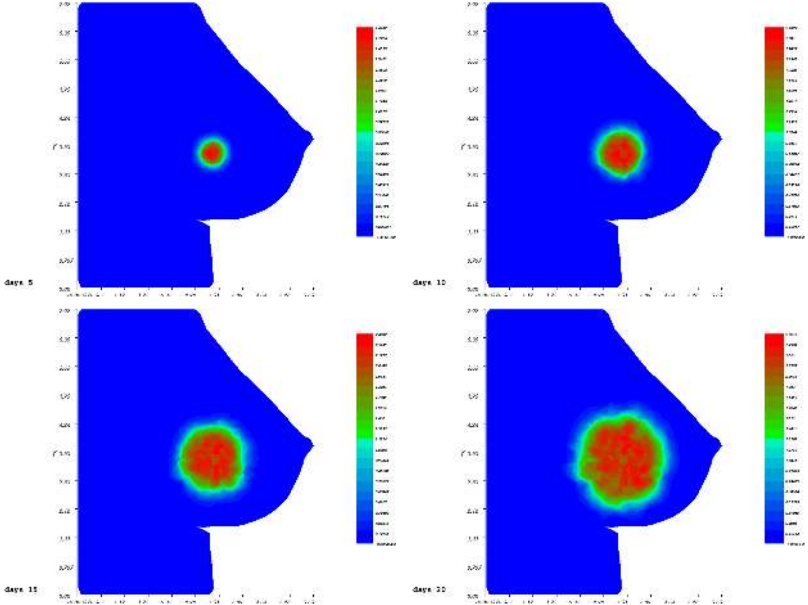
PART 2-4. Preliminary results

The influence of noise on tumor density evolution without treatment on the left and the noise ξ_1 at 20th day on the right.



PART 2-4. Preliminary results

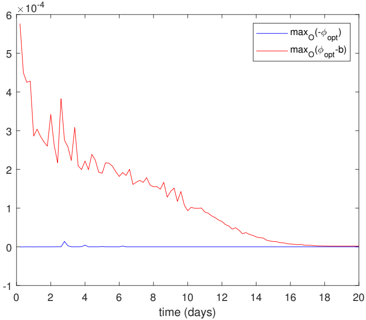
The density of tumor cells $\mathbf{u}$ without treatment *i.e*, $\phi = 0$ in (25) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

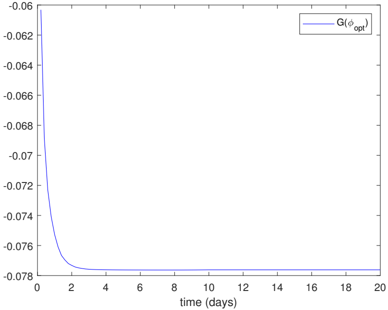


PART 2-4. Preliminary results

The penalised cost parameters
$\lambda = 10^{-2}$, $\beta_1 = 10^4$, $\beta_2 = 3 \cdot 10^5$, $\beta_3 = 10^2$

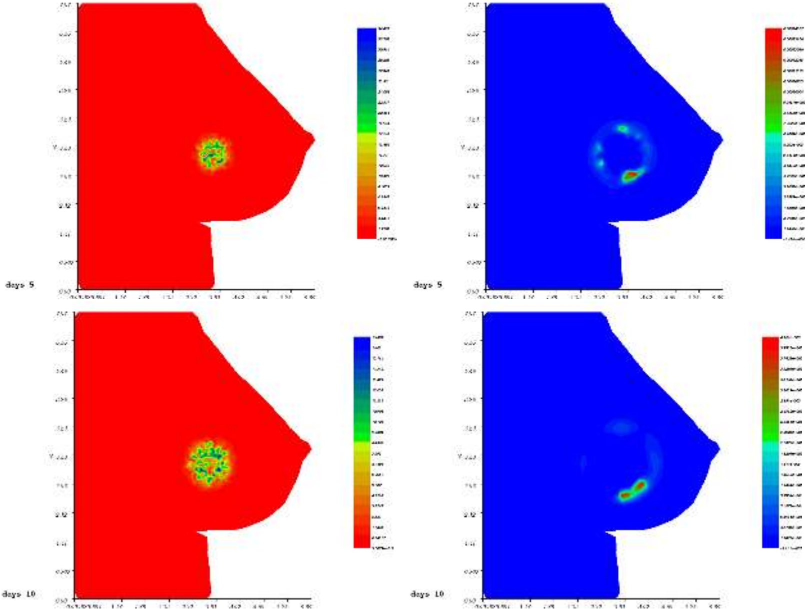
Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following



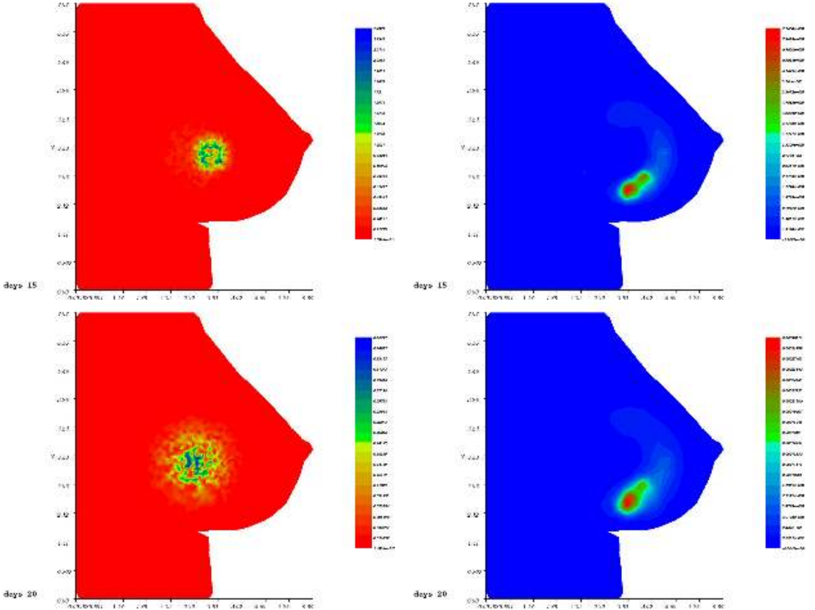


PART 2-4. Preliminary results

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

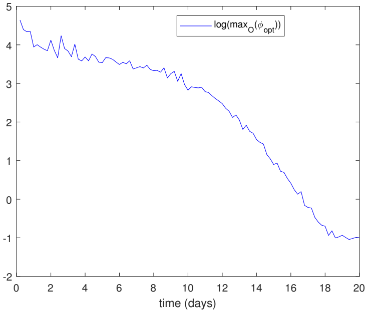


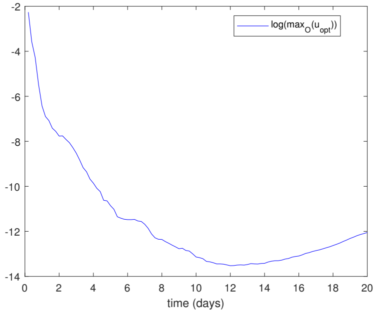
PART 2-4. Preliminary results



PART 2-4. Preliminary results

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





PART 2-5. Treatment of lung tumors

$$\mathrm{D} = \mathrm{D}_0\, e^{-\gamma_0 \sigma}; \quad \sigma = \sigma_0\, e^{-\delta_0 |\mathrm{x}|^2 t}; \quad \mathrm{K}_1 = k_1\, e^{-\delta_0 |\mathrm{x}|^2 t}; \quad \mathrm{K}_2 = \theta^2 \mathrm{K}_1; \quad \mathbf{a}(\mathrm{x}) = 0;$$

$$\mathbf{b}(\mathrm{x}) = b_0\, e^{-b_1[(x-x_1)^2+(y-y_1)^2]}; \quad \mathrm{K}_3 = k_3 \mathrm{K}_{1\Gamma}; \quad \mathrm{D}_0 = d_0 \eta_1; \quad \xi = \|u_0\|^2_{\mathrm{L}^2(\Omega)}.$$

The discretization and mesh informations are the following

$\Delta t = 0.2$ days (the time step) ; $\mathrm{x} = (x, y); \Delta x = \Delta y = 1mm$;

$n_e = 1492$ (number mesh elements) ; $n_p = 821$ (number mesh points) ;

$(x_0, y_0) = \Big(1.34, 3.8\Big)$ (the tumor center) ; $(x_1, y_1) = (1.3, 3.5)$.

Parameters	d_0	δ_0	k_1	α_0	σ_0
Values	$0.2\ mm^2/d$	$5.55 \cdot 10^{-5}/mm^2 \cdot d$	$0.44/d$	$1/\mu\mathrm{M} \cdot d$	25 mmHg

γ_0	k_3	T	ς_1	ς_2	ς_3	ς_4	θ	λ	β_1	β_2	β_3
$1/mm$Hg	$3\ mm$	$20\ d$	0.7	50	2	10^{-4}	2	10^{-2}	10^3	$3 \cdot 10^4$	$5 \cdot 10^6$

b_0	b_1
$100\ \mu\mathrm{M}$	$11.11mm^2$

PART 2-5. Treatment of lung tumors

(1) Case of tumor in the left lung

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} \text{ in } \Omega. \tag{29}$$

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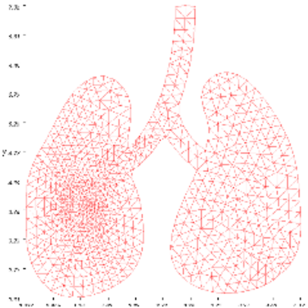
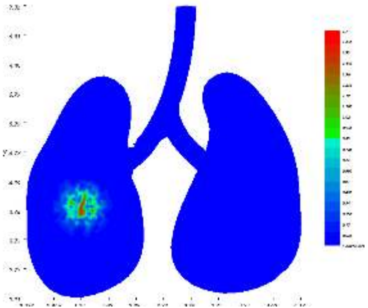
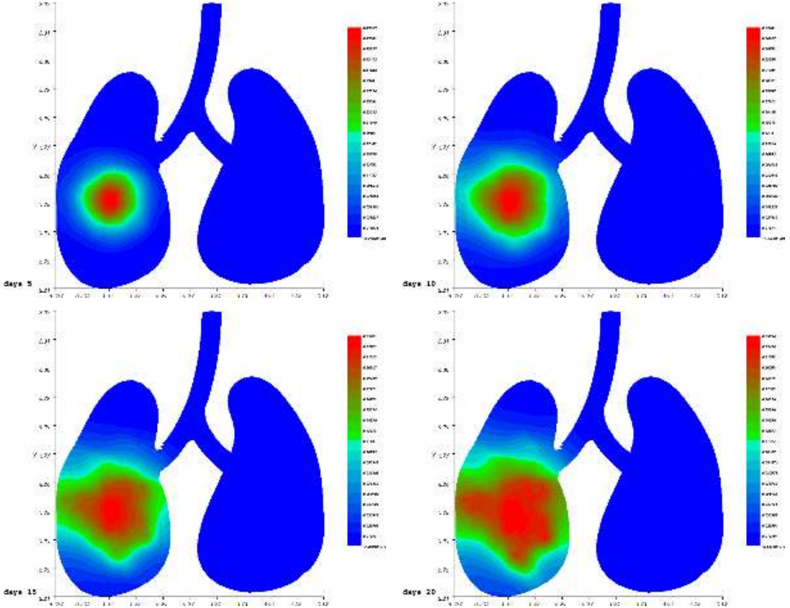




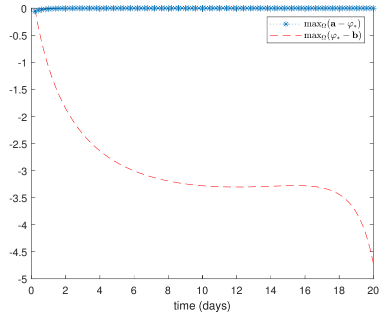
PART 2-5. Treatment of lung tumors

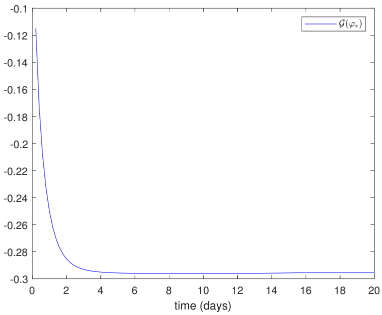
The density of tumor cells $\mathbf{u}$ without treatment $i.e$, $\phi = 0$ in (1) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



PART 2-5. Treatment of lung tumors

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

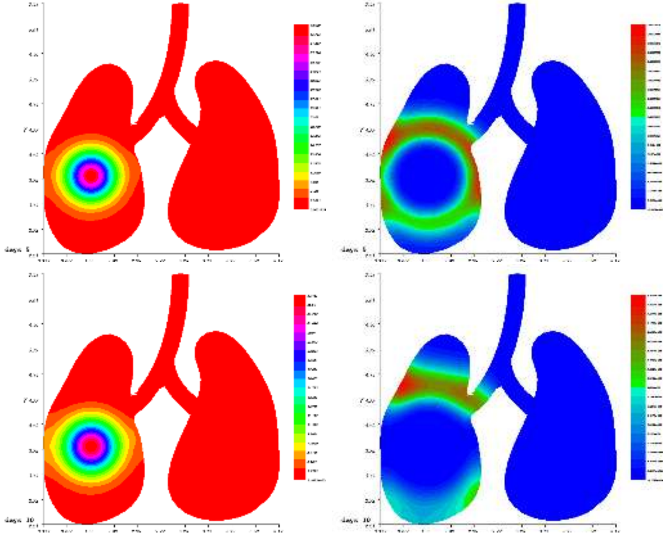




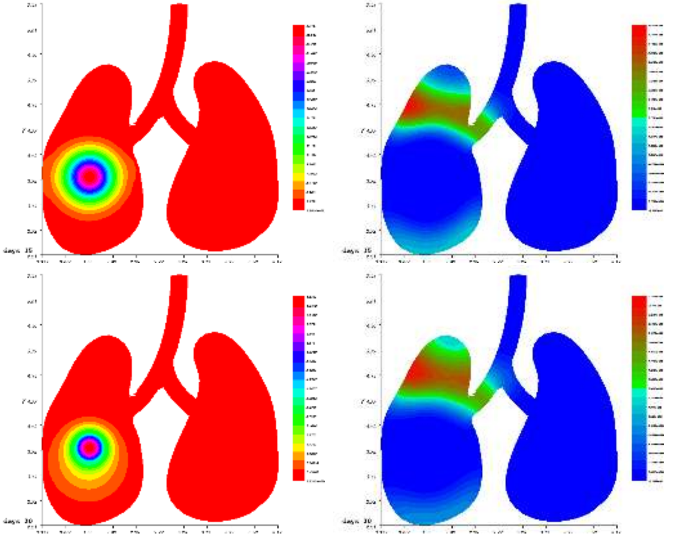
Time (t)	5th day	10th day	15th day	20th day
$\max_\Omega \mathbf{u}_*$	0.00045	$6.10 \cdot 10^{-5}$	$3.54 \cdot 10^{-5}$	$2.72 \cdot 10^{-5}$
$\min_\Omega \mathbf{u}_*$	$1.55 \cdot 10^{-7}$	$4.17 \cdot 10^{-10}$	$8.78 \cdot 10^{-11}$	$1.14 \cdot 10^{-12}$

PART 2-5. Treatment of lung tumors

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

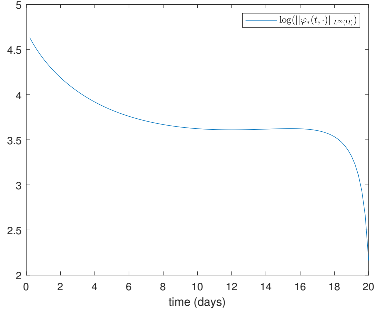


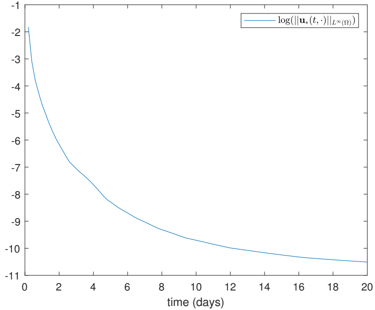
PART 2-5. Treatment of lung tumors



PART 2-5. Treatment of lung tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





PART 2-5. Treatment of lung tumors

(2) Case of tumor in the right lung

$$u_0(\mathbf{x}) = \eta_1 e^{-\frac{1}{\varepsilon}[(x-x_0)^2+(y-y_0)^2]} \text{ in } \Omega. \tag{30}$$

Mesh data and the parameters

(x_0, y_0)	(x_1, y_1)	ε	n_e	n_p	λ	β_1	β_2	β_3
$(3.8, 4.5)$	$(1.3, 3.5)$	0.1	1444	797	$5 \cdot 10^{-2}$	$2 \cdot 10^{4}$	10^{5}	10^{4}

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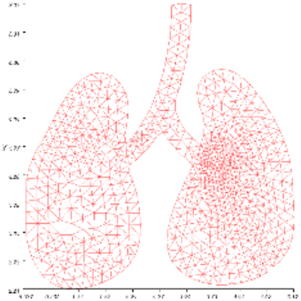
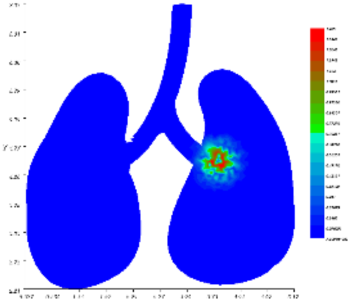
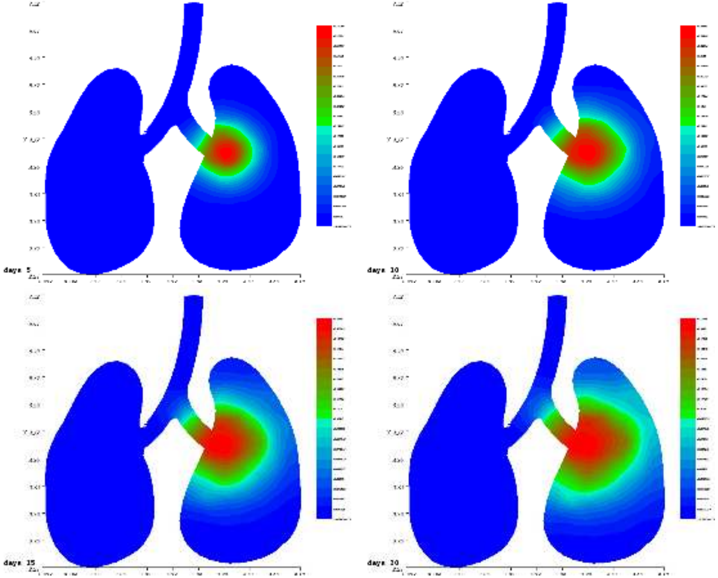




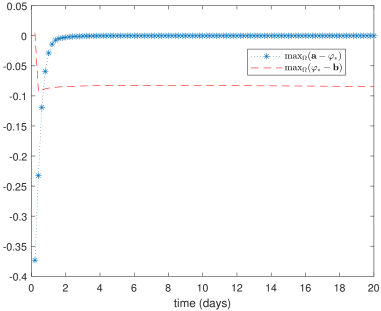
PART 2-5. Treatment of lung tumors

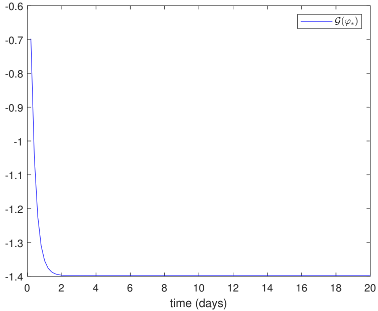
The density of tumor cells $\mathbf{u}$ without treatment *i.e*, $\phi = 0$ in (1) at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days



PART 2-5. Treatment of lung tumors

Constraints satisfaction of the optimal control ϕ_* and optimal state $\mathbf{u}_*$ in the following Figurescf

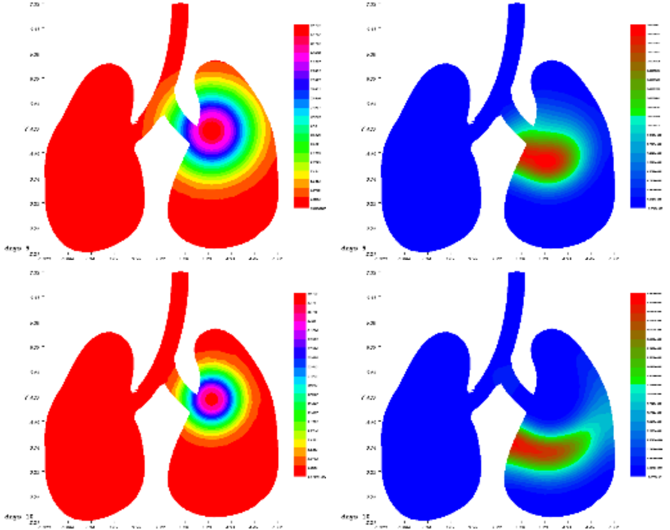




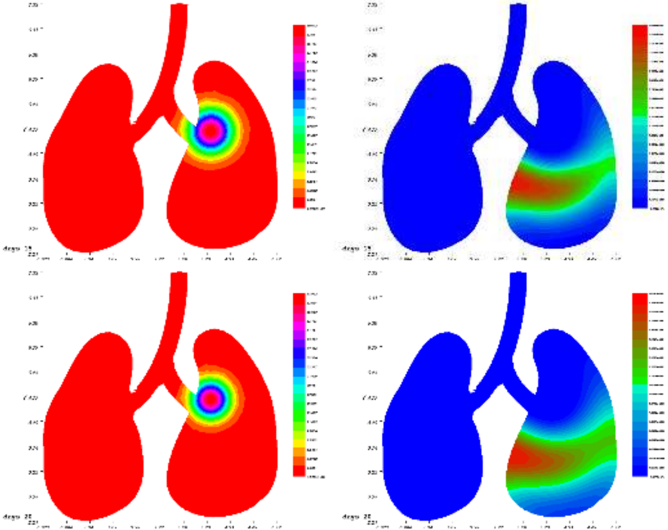
Time (t)	5th day	10th day	15th day	20th day
$\max_\Omega \mathbf{u}_*$	0.00067	$4.49 \cdot 10^{-5}$	$2.65 \cdot 10^{-5}$	$2.09 \cdot 10^{-5}$
$\min_\Omega \mathbf{u}_*$	$1.17 \cdot 10^{-7}$	$5.37 \cdot 10^{-11}$	$6.83 \cdot 10^{-12}$	$1.037 \cdot 10^{-12}$

PART 2-5. Treatment of lung tumors

The optimal drug concentration ϕ_* and optimal tumor density $\mathbf{u}_*$ at $5^{th}, 10^{th}, 15^{th}, 20^{th}$ days

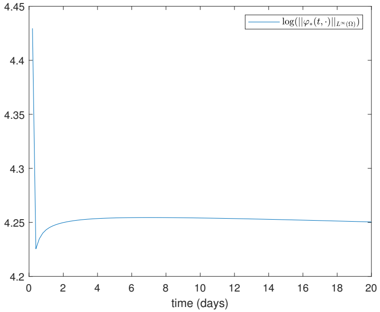


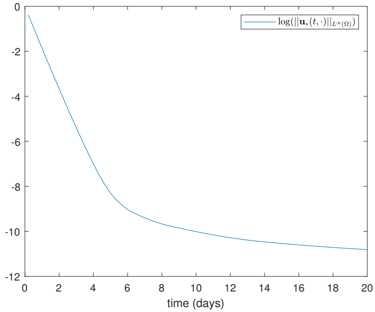
PART 2-5. Treatment of lung tumors



PART 2-5. Treatment of lung tumors

Evolution of the maximum value of optimal control ϕ^* and state $\mathbf{u}^*$ in the logarithm scale





Outline of the talk

Conclusion and future work

What is done ?

- New methodology for the traitement of breast and lung tumors.
- The simulation results prove the importance of the constraints and are promising.

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Open problem!!!

Study of discontinuous treatment strategies (chemotherapy cure).

Thank you for your attention!!!